SignLingo

AI-powered sign language trainer

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1. Project Description

A web application powered by computer vision to help individuals learn sign language on their own,the application will teach new signs to the learner and ask them to repeat them, it would then use computer vision to analyze and give feedback to the learner on what they got right/wrong, eventually building to communicating entire sentences and thoughts.

2. Problem identification

2.1- What problem does this project aim solves ?

Learning sign language on one's own is incredibly difficult due to the lack of feedback the learner receives ,this contributes to driving off potential learners and make an already rare humanitarian skill even more rarer among hearing folk

2.2- Why is this problem important ?

currently there are over 410 million deaf people around the world that rely on sign language As their main form of day to day Communication, in contract there Are only around 12~30 million Hearing individuals that know sign Language, making learning sign Language more accessible aims to bridge this gap seen in fig(1-1)

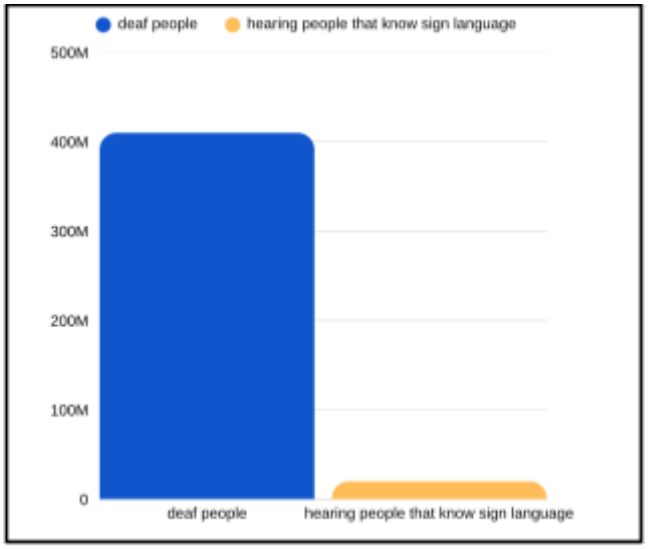


Fig 1-1

2.3- **Who experiences this problem ?**

Any individual who wants to learn sign language for communicating with loved ones or working with deaf people and is limited to learning it on their own for any number of reasons like limited time schedule, difficulty finding a coach or just for convenience.

2.4- **Current solutions**

through a quick search over the internet i found current solutions to this problem to be limited and often none tailored to the specific problem, these include:

- Learning from online videos (no feedback)
- Learning from an llm (not tailored for the problem)
- Asking a human to teach you (binds you to a schedule)

3. Proposed solution

3.1- general system workflow

At it’s core the system works by showing the user signs gestures through their browser and asks them to replicate it in front of a camera, the system then uses computer vision specifically through hand landmark detection , the system then uses an LMM-agent to analyze the the user’s signed gesture and compare it to a database of how the sign is actually performed, the system would then give the user feedback of what they did right and what they got wrong, this feedback might look something like this “position of both hands is correct, but try extending the right index finger more”, the system would begin with teaching the user words and gradually transition into complete sentences, for more details about the actual technologies see chapter(5).

3.2- why is this solution better than existing ones ?

The system combines the convenience of learning sign language online with the feedback analysis of asking a human to teach you so you get the best of both worlds, you can learn at you own pace at your own time and not have to worry about not doing the gestures right thanks to the instant feedback you get.

4. Business analysis

4.1- target customer

Individuals that want to learn sign language

4.2- why would customers pay for the service ?

The service offers its customers 2 main advantages over existing methods those being:

1- reduced cost:

Hiring a professional to teach you is highly costly, learning promises similar results for a much lower fee.

2- time freedom:

The user is not tied to a schedule and is free to learn at their own pace.

4.3 business model

For this kind of application, similar business products exist for spoken languages, we can look at said products and take inspiration from how they do business, one product i will focus on is **Duolingo** as it is essentially solves the exact business problem for spoken language, with that in mind here is the proposed business model:

1- limited free plan with ads and daily use limits

This allows the user to try the product and gets them hooked on the service it offers

2- full access plan through a monthly subscription

A factor that has to be considered when deciding on the final price of the plan is it has to be lower than the cost of hiring a human to teach you.

4.4 Competitors

I have found 2 online services that aim to tackle the problem of learning sign language online however none seem to utilise the power of computer vision those are,

1. **Signlanguage101** at www.signlanguage101.com

   Which offer paid full courses consisting of recorded videos of sign language expert trainers, it also features an ai powered sign language translator, but not for the learning side

2. **HandSpeak** at www.handspeak.com

Which offer something closer of what my product aims to in the form of short recorded clips of individual words and sentences being signed

4.5 Competitors limitation & what the product does different

As you can see, both examples i gave above are limited in one area: **feedback,** said limitation come from the nature of only relying on pre-recorded videos to learn a skill that needs a trained eye to know what is right and wrong such as sign language which often leads to the user not realizing they are making a mistake, that is exactly what this product aims to improve, by added a **feedback-loop** to the process of learning sign language online, achieving this through computer vision models, and the use of llm-agents to better communicate thoughts.

5. Technical analysis

5.1 type of computer vision problem

At its code this is a object recognition/classification problem in the form of detecting human hands in an image, and recognizing the sign or gesture formed by the hand, classifying signs into distinct classes

5.2 existing data

For this product, some datasets -although limited- do already exist, on the other hand to achieve the product goal of a full sign language vocabulary, a lot of other data will also have to be gathered examples of existing datasets are:

1- American Sign Language Dataset on Kagle

Consists of 2515 samples of raw images split between 36 classes for each letter and digit in the english alphabet, however there are no samples of words or expressions in this dataset

2- ASL Alphabet Hand Landmarks on Kagle

Consists of 10,508 balanced samples of 21 hand landmarks built on the **Google-MediaPipe-Hand-Landmarker**(see section 5.4) model bundle,it splits data across 26 classes for the english alphabet letters, also no classes for words or expressions.

5.3 collecting specialised data

As exciting datasets are limited to only letters, we would need to collect specialised data for complex vocabulary and expressions, after reviewing options it seems there are 3 methods this can be done, regardless of which way is chosen, these methods won't provide clean feature-ready datasets and a lot of feature extraction would have to be done, here are the mentioned three methods:

1- work directly with experts

hire a small team of sign language experts to provide the required data to feed to the model, this approach would be the most reliable although it would also cost the most.

2- public sources

This is the opposite approach to working with professionals as it depends on gathering data from publicly available recorded sign language videos and dictionaries on the internet, this way this would be the least costly approach but it also depends on said data being accurate and clean, which it may not be. Public dictionaries do exist like www.signasl.org, (one approach to automate this process is to use an LLM-agent to browse and query words into those dictionaries and collect the videos)

3- buy from dataset marketplaces

This strikes a form of middle ground between the 2 approaches mentioned, as these datasets are often more reliable than working with public sources, and are also cheaper than hiring experts, marketplaces like www.talksign.co/marketplace offer multiple paid clip-datasets like the **ASL Conversational Corpus** which consists of 240 hours of expert-reviewed clips for conversational signs, multiple other data sets for different sign languages and specialization exist.

5.4 Models and algorithms

For solving this type of computer vision problem, it seems to me that two approaches exist, those are:

1- pixel-based approach

Which makes use of raw CNN-to-FC models to analyze the image in the classical filter/kernel based way to extract features related to the gesture of the hand and feed said features to a fully connected classifiers, to determine the class that most fit the the images of human hands, models like **ResNet50**, **INSEPTION** or **VGG16,** could be used for this approach.

2- landmark-based approach (my pick)

The idea of landmark based classification is -in my opinion- best described as follows, instead of relying on raw feature maps gathered from raw pixel values to train the fully connected classifier, we can use a more sophisticated system to extract positional landmarks from the image of the hand, and train the classifier on said landmarks, one promising solution i found is **mediapipe-hands** provided by the mediapipe framework, it uses two models: **Palm Detection Model** and **Hand Landmark Model** to detect 21 positional landmark points across the hand in (x, y) coordinates, fig(5-1) shows exactly what those points are for further information the solution's documentation is at https://mediapipe.readthedocs.io/en/latest/solutions/hands.html#palm-detection-model,

here are some of the solution's important hyperparameters :

- **Max_num_hands :** the maximum number of hand the model is allowed to detect in an image, for this task this should be set to 2

- **Min_detection_confidence :** the confidence threshold on which the solution decides if the image contained a hand or not, default is 0.5

- **Static_image_mode :**If set to **false**, the solution treats the input images as a video stream. It will try to detect hands in the first input images, and upon a successful detection further localizes the hand landmarks. In subsequent images, once all **<max_num_hands>** hands are detected and the corresponding hand landmarks are localized, it simply tracks those landmarks without invoking another detection until it loses track of any of the hands. This reduces latency and is ideal for processing video frames. If set to **true**, hand detection runs on every input image.

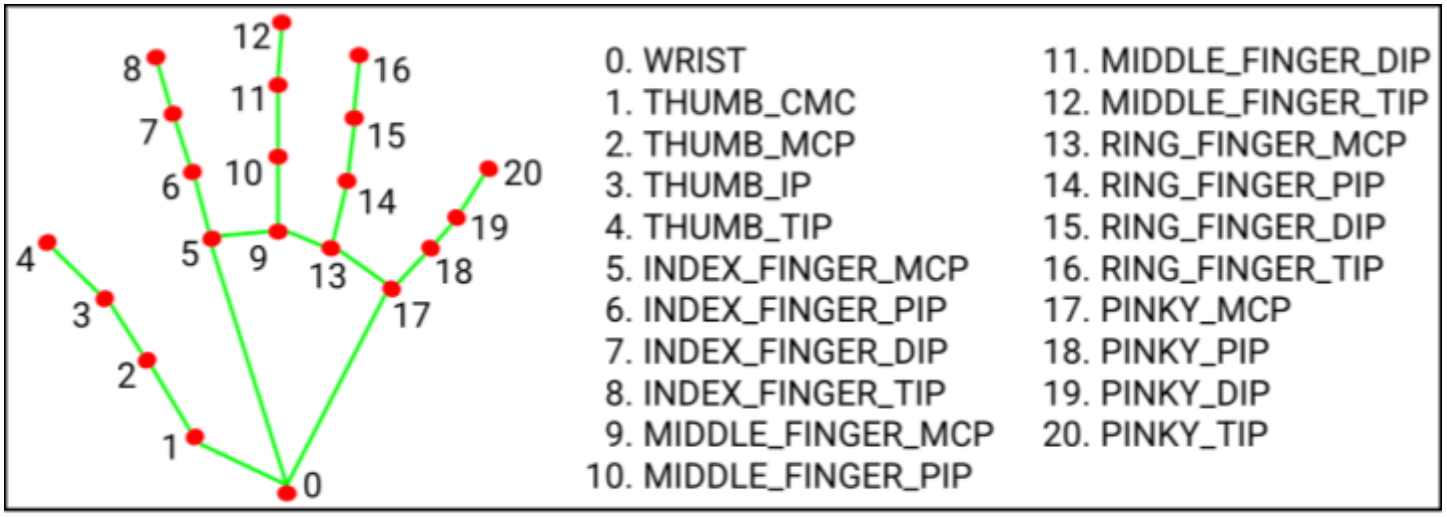


Fig 5-1

5.5 tech stack

(As this is an ai hackathon i will focus on the ai side, I also don't have enough experience in tracks like frontend or cloud to list a tech stack for them, and i dont want to list technologies i dont understand)

1. **Python** (core language)
2. **Numpy** (mathematical operations)
3. **Pandas** (data management)
4. **pyTorch** (deep learning utilities)
5. **openCV** (computer vision utilities)
6. **huggingface/media-pipe** (handlandmarking approach models)
7. **CNN** (pixel approach models)
8. **[in the future] transformers** (core llm support)
9. **[in the future] langchain/langgraph** (agentic integration)
10. **[in the future] RAG** (controlled agent feedback)

6. Feasibility analysis

6.1 technical possibility:

From a technical side, this project shows a simple linear workflow for each track, with a small team of **ai, backend, frontend, and cloud** developers, this project is very much possible to complete within a month, the main reason I am confident in saying this is because platforms like **Duolingo** already exist, the team can take inspiration from such platforms, we don't have to “reinvent the wheel”

6.2 technical challenges

1- scalability

As is tradition with almost all cs projects, in my opinion the biggest challenge that faces this project is **scaling**, i believe the core idea of the project is sound, but as the project transitions into an actual app, that should support full vocabularies, planning how the project will scale will be the single most difficult task and the one that either makes or breaks the future of the project, as a project that doesn't scale is practically useless

2. Data availability

I have already discussed this in sections (5.2) and (5.3), to recap, most of the data of this project would either be collected from scratch, bought, or gathered of the internet, with each approach having upsides and downsides.

6.3 hardware required

At its core this is a software project, model training would ideally be done on cloud super computers, hosting would also rely on cloud providers like aws

6.4 Minimum Viable Product

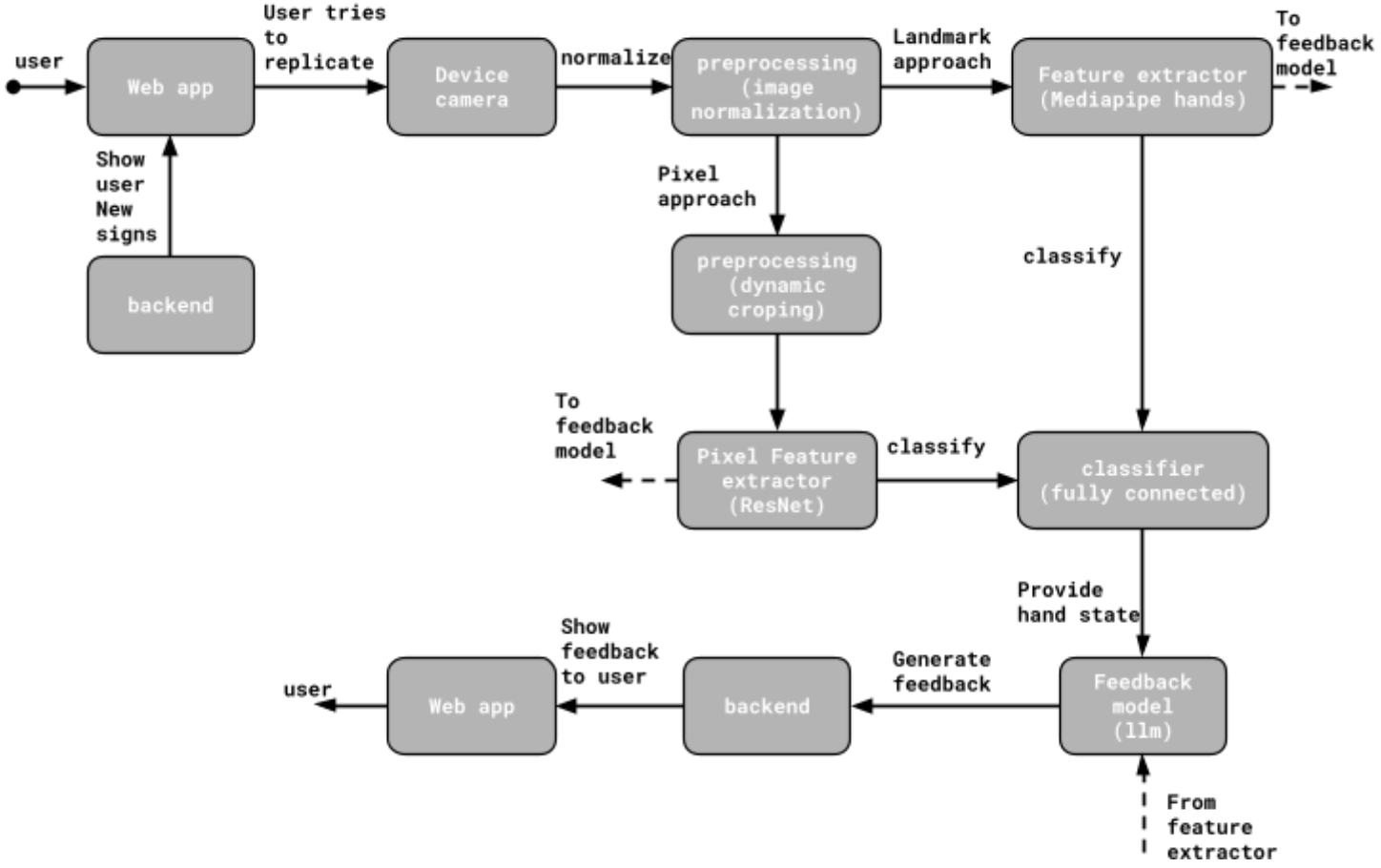
I have actually waited to write this part until i build my project as this is the idea i will doing for it anyways, and seeing what i managed to achieve alone in 4 and half days, i believe i have a pretty good idea what a small team may be able to build in about 2 months.

1. **What i build alone in 4 days:**
   - A signed-asl-letter classifier that flags whether you sign the each letter right or not, the models are as bare bones as they come,the size training data is laughable ,only one hand support,only letters support, no smart feedback, no gui or backend, **but** it does prove the point **“computer vision can be used to solve this task”**

2. **what a small team would build in 2 months**
    - Something closer to an actual version-0, with backend and user interface, much more dedicated models trained on respectable sized data, by this pace it **should** support
        - Bare bones business logic
        - Support for complex signs with both hands
        - Support for a good size vocab of ASL (<= 100 words)
        - Support for actual user feedback
        - Support for quick vocab addition
        - An actual graphical user interface
        - Backend with clear plans for hosting

    - If the previous state was met with spare time some good **optional** improvements would be
        - Support for agentic based feedback
        - Support for a user positive reinforcement loop that reward frequent app usage
        - A larger vocab

7. Proposed Final Project Architecture



8. Research Sources

8.1 technical sources

- **mediapipe/hands documentation :**
  https://mediapipe.readthedocs.io/en/latest/solutions/hands.html
- **Pytorch/resnet documentation :**
  https://docs.pytorch.org/vision/main/models/resnet.html
- **openCv documentation :**
  https://docs.opencv.org/5.0/

8.2 dataset sources

- **Kaggel ASL landmarks data**
  https://www.kaggle.com/datasets/borisgraudt/asl-alphabet-hand-landmarks
- **Signasl ASL dictionary**
  www.signasl.org
- **Talksign sign language data marketplace**
  www.talksign.co/marketplace

8.3 Business sources

- **Paper on Duolingo's business model**
  https://businessmodelanalyst.com/duolingo-business-model/
- **Paper on the market of sign language learning apps**
  https://dataintelo.com/report/sign-language-learning-apps-market